Paper1: Perturbation-based gene regulatory network inference to unravel oncogenic mechanisms Daniel Morgan1

The gene regulatory network (GRN) of human cells encodes mechanisms to ensure proper functioning. However, if this GRN is dysregulated, the cell may enter into a disease state such as cancer. Understanding the GRN as a system can therefore help identify novel mechanisms underlying disease, which can lead to new therapies. To deduce regulatory interactions relevant to cancer, we applied a recent computational inference framework to data from perturbation experiments in squamous carcinoma cell line A431. GRNs were inferred using several methods, and the false discovery rate was controlled by the NestBoot framework. We developed a novel approach to assess the predictiveness of inferred GRNs against validation data, despite the lack of a gold standard. The best GRN was significantly more predictive than the null model, both in cross-validated benchmarks and for an independent dataset of the same genes under a different perturbation design. The inferred GRN captures many known regulatory interactions central to cancer-relevant processes in addition to predicting many novel interactions, some of which were experimentally validated, thus providing mechanistic insights that are useful for future cancer research.

Cancer can be seen as an altered state of the regulatory systems that control cell proliferation and cell death. Such systems are generally not sensitive to individual gene malfunctions, but an aggregation of aberrations can lead to sufficient dysregulation to cause cancer. Reliable models of these regulatory interactions would offer insight into key mechanistic alterations for therapeutic targeting. Cancer subtype-specific gene regulatory networks (GRN) encode intracellular dynamics1 , and offer understanding into the functional changes driving disease development.

Methods using data from known and systematic perturbations have shown greater accuracy among inference techniques since more information is available to determine regulatory causal mechanisms in the system

GRN inference has proven its value to unravel novel regulatory links of biological significance. For instance, ARACNe was applied to gene expression profiles to predict a glioma-specific GRN, revealing that C/EBPbeta and STAT3 are master regulators of mesenchymal transformation, which was validated experimentally6 . In another study, eight key genes were knocked down by siRNA, and the gene expression together with prior knowledge were used to infer a GRN network in the RAS pathway with good validation performance7 . A large number of GRN inference algorithms exist. In a survey by the DREAM5 project8 it was shown on simulated and E. coli data that some methods performed better than random predictions. However, many methods did not outperform random prediction, and on yeast data no method performed much better than random selection. Since this study, the community has developed methods for integrating various priors: literature/ database, ATAC-seq, DNase I hypersensitive sites, ChIP-Seq, or proteomics data to increase information about the system.
In this study, we deployed perturbations through siRNA gene knockdown of each gene in our literaturecurated set of cancer-related regulator genes, each followed by transcriptomics measurements of all genes, in order to measure the global influence of each individual gene. Knockdown experiments are more informative about the system than irreversible and complete knockout which may cause drastic rewiring of the underlying network into another system entirely.

“In this way a selected set of 40 genes relevant to cancer were perturbed, and the transcriptomic response data were used to construct a model of underlying regulatory interactions. We inferred these interactions by relating the effect of the gene perturbations to the expression of the readout genes, using three GRN inference algorithms well suited for employing our linear model and perturbation design: LASSO, LSCO, and TLSCO. The regulatory interactions inferred by these methods are not limited to direct physical interactions, but should be seen as regulatory influences, which may be indirect via genes that are not modeled because they were not measured. ”

A drawback of all GRN inference algorithms is that they generally produce erroneous GRNs if the noise level is high15,16. To ensure inference of reliable GRNs, we employed NestBoot19, a recent framework implementing nested bootstrapping, wrapped around any individual GRN inference method to better account for sample variation and noise. Contained within the GeneSPIDER package16, NestBoot generates bootstrap support distributions for links inferred from measured as well as shuffled data20, and minimizes false links by comparing them. This way Nestboot is able to discard links even if they have high bootstrap support, if they also have this in the null distribution. NestBoot has been shown to give substantially increased inference accuracy across both synthetic and experimental datasets when compared to the methods in their native implementation

Methods

- Knockdown data collection:
  A set of genes was assembled from different pathways and complexes, each interacting to some degree with the oncogene MYC21 (Tables S2 and S6). Each readout gene was perturbed in the human squamous carcinoma cell line A431 via transfection with short interfering RNAs (siRNAs). We then harvested, purified and prepared libraries using the Ambion Library Construction Kit22. A precise record of perturbations is key to modeling (next section). In order to minimize siRNA off-target effects, two to three siRNAs were used per target (Table S4), which were then averaged to purify the effects of the targeted siRNA perturbation. Cells were collected 72 h after siRNA knockdown and washed of Phosphate-Buffer Solution (PBS), and lysed using CelluLyser23. Cell counts were calculated using the resazurin fluorescence assay. Since no endogenous gene can be assumed to be free of MYC regulation, which is thought to be a universal transcriptional amplifier24, a spike-in RNA transcript was added to each sample to act as a reference gene for the quantitative polymerase chain reaction (qPCR) analysis, added in proportion to the cell count before RNA isolation. It consisted of a 1,000-base sequence with a 5′ cap and a polyA tail. This was only used for normalization of mRNA level across samples25. Negative controls were included of siRNA not mapping to human genes, as well as an untreated control absent of any siRNA. The cDNA was prepared from the RNA and preamplified in preparation for the high-throughput qPCR screening. Finally, the transcript profiles with respect to the 40 genes were determined with TaqMan qPCR assays (Table S5) using Fluidigm Biomark 96×96 Dynamic Array integrated fluidic circuits. Raw qPCR output was processed with the ddct R package26 into log transformed fold changes relative to the experimental controls. Three experimental replicates were made per targeted perturbation and five outlying replicates were discarded due to clear machine read error, thus the dataset is composed of 40 genes (N) and 115 samples. Including all controls, a total of 18,432 qPCRs were performed on 192 samples. Two technical replicates were performed to ensure minimal machine error. They generated very similar values up to 25 qPCR cycles (Fig. S1). Experimental validation of individual interactions was performed on GTML2 brain tumor cells, which were cultured in serum-free stem cell medium as previously described27 and treated for 2 h with DMSO or JQ1 (500 nM). RNA was purified using the RNeasy Kit (Qiagen). RNA sequencing was performed using the

Ion Proton System for Next-Generation Sequencing at NGI, SciLifeLab, Uppsala Biomedical Center (BMC), Sweden. All treatment conditions were performed in triplicates. All RNA sequence reads were processed and the differentially expressed genes were analyzed as previously described27. An additional validation was based on a gene expression data set comprising DMSO- and JQ1-treated glioma cell lines, which we had previously published28. Specifically, in this study we were able to distinguish between JQ1-resistant and JQ1-sensitive human adult highgrade glioma cell lines. From the four cell lines with available AmpliSeq expression data, only one (U3056) was characterized as JQ1-sensitive and expressing high MYC levels, and was accordingly selected to investigate the interaction between BRD4 and CCNB1. Expression data for the U3056 cell line was downloaded from the Gene Expression Omnibus (GSE138942) and comprised 6 h DMSO and JQ1 treatments in triplicates each.

- GRN inference:
  The fold change is calculated in comparison to the spike-in for all knockdown experiments. It is used in combination with the collective experimental design matrix (describing the location of perturbed and readout genes) to determine the GRN, i.e. the interaction matrix A, of regulatory effects from gene j to i in element aij. We use a linear ODE model, similar to17,29,

  which simplifies to $$Y = -A^{\dagger}(P - F) + E$$

  where Y is an expression matrix of calculated fold changes, with N genes (rows) and M experiments (columns). In Eq. 1, P is the design matrix if we solve for A†, where the Moore–Penrose generalized inverse, denoted †, is used throughout in place of the inverse due to computational intractability wherein sparse GRNs might be rank deficient. However as we want to solve for A and not A†, we reformulate Eq. 1 to a traditional regression problem on errors-in-variables form, −(P − F) = A(Y − E). The error in Y and P are represented as measurement error E and process error F, respectively, as defined in Table S1. F is used as an estimate of the variation in the perturbation, e.g. siRNA efficiency or environment, while E is used as an estimate of the variation inherent to the cells' expression as well as error in plate reading26. Three methods are employed to perform model selection and parameter estimation simultaneously. LSCO (least squares with a cut off to produce variably sparse networks)30 was chosen for its resemblance to the standard ordinary least squares method, LASSO (least absolute shrinkage and selection operator)31 was chosen for its proven ability to find the sparse solution with minimum errors, and TLSCO (total least squares32 with the same sparsity inducing cut off as LSCO) for its ability to model error in both the dependent and independent variables15. Each method is encapsulated within the nested bootstrapping framework to estimate the linear model in an accurate and reproducible manner by limiting false discovery rate (FDR), in their native configuration
- GRN validation without gold standard. To evaluate the goodness-of-fit of the inferred network in a prediction error framework one needs to balance the measurement and process errors. This optimization occurs during the leave one out procedure (Algorithm 1), using the CVX convex optimization package (v1.22)33 for MATLAB, where the left out gene (g) is expressed as a linear combination of the other experiments (crossvalidation). The aim of this procedure is to equally balance the measurement and process errors when predicting the left out gene under cross-validation.

Algorithm 1 : *Balanced Fitting of Errors* with cross validation (BFECV) to balance the fit error equally between the measured output (Y) and process input perturbation design matrix (P).

1. For each perturbed gene *g*:
   a. Remove all experiments (columns) where g is perturbed from $Y, P \rightarrow Y_{!g}, P_{!g}$
   b. While $|F_{rel} - E_{rel}| > t$
      i. Minimize $\left|\left|E_{!g}\right|\right|_{Fro} + \left|\left|F_{!g}\right|\right|_{Fro}$ subject to

$$A\,(Y_{!g} - E_{!g}) = -(P_{!g} - F_{!g})$$

$$\left|\left|E_{!g}\right|\right|_{Fro} <= \frac{1}{\chi^2(\alpha, NM - number\ of\ links\ in\ A)}$$

$$\left|\left|F_{!g}\right|\right|_{Fro} <= w\,\|A\,Y_{!g}\|_{Fro}$$

      ii. $E_{rel} = \frac{\|E_{!g}\|^2_{Fro}}{\|Y_{!g}\|^2_{Fro}}$ and $F_{rel} = \frac{\|F_{!g}\|^2_{Fro}}{\|P_{!g}\|^2_{Fro}}$

      iii. Use bisectioning to find the value of *w* that minimises $|F_{rel} - E_{rel}|$

   c. Estimate process noise of gene *g* by $F_g = F_{!g}\,(P^{\dagger}_{!g}\,P_g)$

   d. $wRSSP_g = \frac{\left|\left|\widehat{P}_g - P_g\right|\right|^2}{var(P)}$, where $\widehat{P}_g = P_g - F_g$

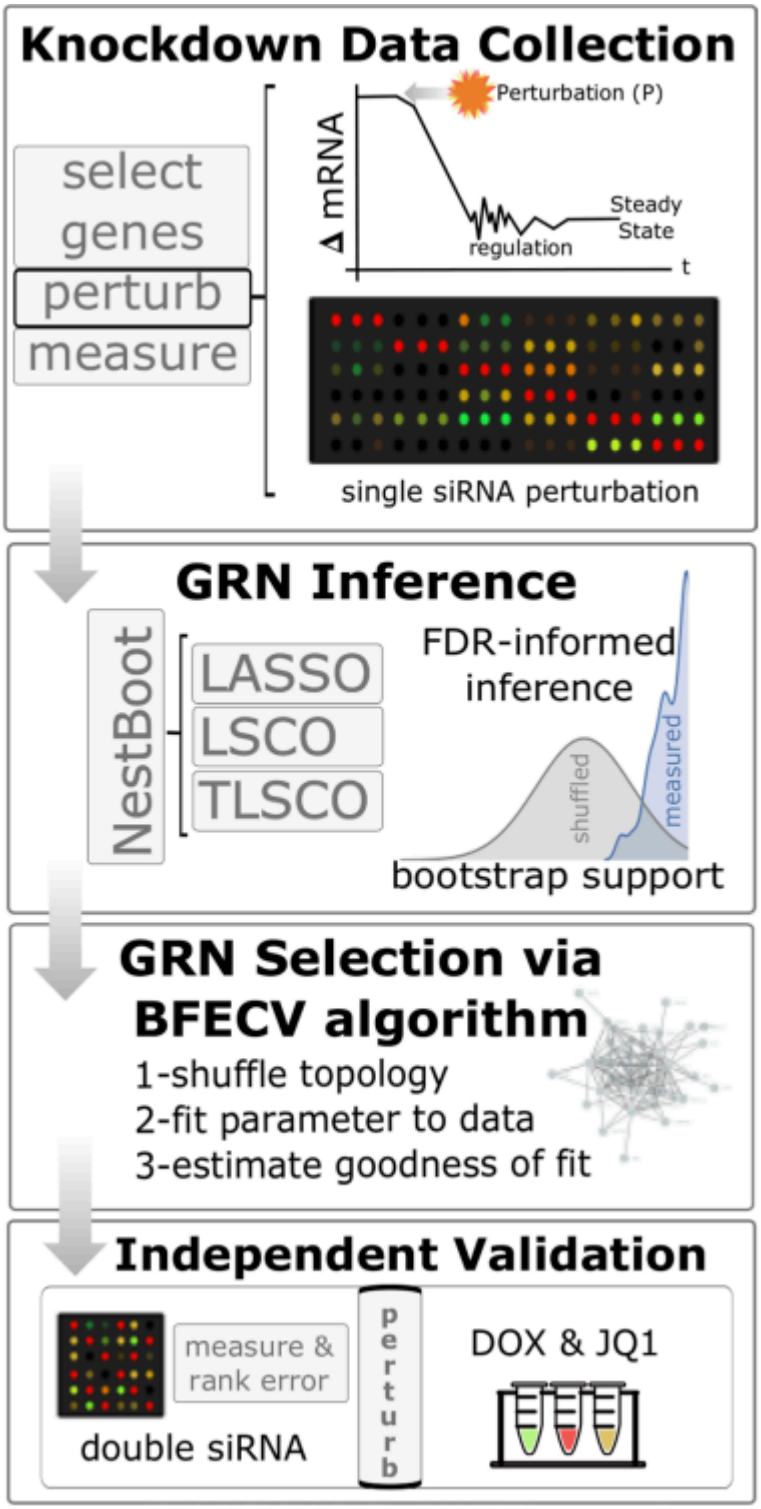
   e. $wRSSY_g = \frac{\left|\left|\widehat{Y}_g - Y_g\right|\right|^2}{var(Y)}$, where $\widehat{Y}_g = -A^{\dagger}\,\widehat{P}_g$

2. $wRSS = \Sigma wRSSY_g + \Sigma wRSSP_g$

In the BalanceFitError algorithm, A contains the inferred GRN structure (topology), with each non-zero value representing a regulatory interaction and each zero a lack of interaction, i.e. pseudo-direct influence. The algorithm estimates each gene's perturbation and response based on the balanced measurement and process errors of all other genes and compares it to the intended perturbation and observed response. Since error is a function of the degrees of freedom of the given matrix, relative error (Erel, Frel) is used to more equally balance these errors. From step (i) to (iii) all matrices have the perturbation experiments of the left out gene g removed and are thus denoted !g. However, A maintains all genes, remaining square throughout, and later the left out experiments can be predicted from the remaining data. This method to evaluate the goodness-of-fit is used on all inferred networks. All inference methods used here have a regularization parameter that determines the number of nonzero parameters in the models, which is varied to span the complete range from empty to full network.

Because our leave out procedure assesses individual gene prediction errors, we assembled null GRN performance distributions by shuffling GRN links and fitting these new networks to the data to create both a standardized and fairly conservative link weight using a constrained least squares (CLS) algorithm30,34. To this end we implement a Monte Carlo sampling method, sampling links to maintain the node in degree and preserving hubs thereby approximating an estimated link null distribution based on the inferred GRN judged to generate conservative and fair null GRNs. For a fair comparison, both the inferred and shuffled GRNs are fit to the original data. However topology and sign are preserved in the GRNs. To obtain a measure of the goodness of fit of both inferred and shuffled GRNs, cross-validation was used to calculate the weighted residual sum of squares (wRSS) of the original training data while balancing the measurement and process errors as described in Algorithm 1. We are able to predict a left out gene in step c and d by expressing it as a linear combination of the other genes. This goodness of fit measure was also made of the inferred GRN's

ability to under cross-validation predict the original data compared to the distribution of prediction errors using shuffled data. The relative error metric comparing measured and shuffled wRSS (Figs. 2, 3, S3, S4) is complemented by R2 values (Fig. S5). Before calculating these, each GRN parameters were modified to ensure that the predicted response remained similarly bounded as the observed gene expression. This was done by performing singular value decomposition of the GRN, setting singular values below a cutoff to zero, and then reconstructing the GRN without the smallest singular values. This GRN was then fit to the training data under cross-validation to generate predicted expression responses in the same way as described above. The cutoff on the minimum singular value was set independently for each GRN to ensure that the predicted expression values were within the range of the measured values. The small singular values generally represent noise if the data is ill-conditioned and removing them reduces the effect of noise. To further verify both predictiveness and generalizability, these GRNs are also applied to a second independent, validation dataset based on the same genes knocked down in pairs, in single replicates. While this data is not used to infer GRNs, we apply the same cross-validation strategy as for the original data to validate the GRNs. This is necessary for parameter fitting, since the process error is different from the single knockdown data. Furthermore, by running the same pipeline we obtain a comparable measure of how well the independent data fits our inferred GRNs, and build null distributions of expected error from shuffled GRNs to examine an inferred GRNs' ability to predict the data.



The perturbation response gene expression data were used for network inference with three methods, LASSO, LSCO, and TLSCO, each run in conjunction with nested bootstrapping. Each GRN inference method was run with varying parameters to produce GRNs in a range of different sparsities. Nested bootstrapping was then used to select the significantly supported links in each GRN, resulting in final sparsities that tend to never exceed 3–5 links/gene even if the natively inferred GRN had 40 links/gene, for example. In order to select the GRN that has the links most likely to exist in reality, we compared how well each inferred GRN's topology fits the data compared to shuffled topologies of the same GRN. The reason for validating the topology rather than the complete GRN including the actual parameter values is that those parameters are optimized to fit the data and therefore no suitable null model exists. Note that by topology we mean the structure of the GRN, i.e. the inferred links and their

sign. For each topology, the parameters were fit to the observed data using cross-validation, and the error was measured as the difference between predicted and observed values of the hold-out samples, after assembling the individual gene predictions into the full predicted matrix. This was done with the novel Balanced Fitting of Errors with cross validation (BEFCV) algorithm (Algorithm 1), which ensures that the error is balanced between sources, i.e. that the error is not merely pushed from the measurement to the process estimation or vice versa. Note that this algorithm is not a GRN inference method on its own but is rather a way to assess the quality of an inferred GRN. Each inferred GRN was shuffled a hundred times, and the data was fit the same way to these topologies to estimate a null distribution of expected inference error. Note that since the parameters of the shuffled topologies are fit to minimize the error, this is a very stringent test of the inferred topology and for a suboptimal GRN one expects some of its shuffled topologies to by chance have lower error. Yet, several of the inferred GRNs greatly outperformed their null model, both when using the original training data (Fig. 2) and the independent validation data with double knockdown design in the same cell line (Fig. S3). We also calculated R2 values to show the proportion of the variation that our GRNs explain

Paper 2: Efficiency of Scale-Free Networks: Error and Attack Tolerance Paolo Crucitti a , Vito Latora b , Massimo Marchiori

Two different kinds of scale-free networks, i.e. networks with power law P(k), are considered: 1) scale-free networks with no local clustering produced by the Barabasi-Albert model and 2) scale-free networks with high clustering properties as in the model by Klemm and Egu´ıluz, and their properties are compared to the properties of random graphs (exponential graphs).

By using as mathematical measures the global and the local efficiency we investigate the effects of errors and attacks both on the global and the local properties of the network. We show that the global efficiency is a better measure than the characteristic path length to describe the response of complex networks to external factors. We find that, at variance with random graphs, scale-free networks display, both on a global and on a local scale, a high degree of error tolerance and an extreme vulnerability to attacks. In fact, the global and the local efficiency are unaffected by the failure of some randomly chosen nodes, though they are extremely sensitive to the removal of the few nodes which play a crucial role in maintaining the network's connectivity.

- **concept of efficiency of a network, which measures how efficiently the information is exchanged over the network. By using the efficiency as a new measure to characterize the network, it has been showed that small-worlds are systems that are both globally and locally efficient. Moreover the description of a network in terms of its efficiency extends the small-world analysis also to unconnected networks and to real systems that are better represented as weighted networks**

Paper 3: Gene perturbation and intervention in probabilistic Boolean networks

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Motivation: A major objective of gene regulatory network modeling, in addition to gaining a deeper understanding of genetic regulation and control, is the development of computational tools for the identification and discovery of potential targets for therapeutic intervention in diseases such as cancer. We consider the general question of the potential effect of individual genes on the global dynamical network behavior, both from the view of random gene perturbation as well as intervention in order to elicit desired network behavior.

Results: Using a recently introduced class of models, called Probabilistic Boolean Networks (PBNs), this paper develops a model for random gene perturbations and derives an explicit formula for the transition probabilities in the new PBN model. This result provides a building block for performing simulations and deriving other results concerning network dynamics. An example is provided to show how the gene perturbation model can be used to compute long-term influences of genes on other genes. Following this, the problem of intervention is addressed via the development of several computational tools based on first-passage times in Markov chains. The consequence is a methodology for finding the best gene with which to intervene in order to most likely achieve desirable network behavior. The ideas are illustrated with several examples in which the goal is to induce the network to transition into a desired state, or set of states. The corresponding issue of avoiding undesirable states is also addressed. Finally, the paper turns to the important problem of assessing the effect of gene perturbations on long-run network behavior. A bound on the steady-state probabilities is derived in terms of the perturbation probability. The result demonstrates that states of the network that are more 'easily reachable' from other states are more stable in the presence of gene perturbations. Consequently, these are hypothesized to correspond to cellular functional states.

Global, systemic, or network perspectives are becoming increasingly important for making progress in our understanding of the biological processes and harnessing this understanding in educated intervention for correcting human diseases. The development of high throughput genomic and proteomic technologies is empowering researchers in the collection of broad-scope gene information. However, it remains a major challenge to digest the massive amounts of information and use it in an intelligent and comprehensive manner. The development of systematic approaches to finding genes for effective therapeutic intervention requires new models and powerful tools for understanding and managing complex genetic networks.
(havent read this particular paper to the end)